How USAFacts' four qualities became something a machine can measure

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The data nonprofit USAFacts set out four qualities that public data should have if it is to be usable by SI systems, the software that now finds, reads and uses data on its own: that it be accessible, understandable, accurate, and open. This work keeps those four qualities as the backbone of its readiness framework, each one a top-level heading left as USAFacts wrote it, under which the detailed measures hang.

To those four it added three areas the guide does not cover: a standing loop that tests and re-tests how well SI systems read the data, the practice of treating every data release the way careful software teams treat a deployment, and a set of concerns particular to official statistics, such as whether a margin of error still travels with a figure once a machine restates it. Each added area rests on named, published standards rather than on assertion, among them the national framework for managing the risks of these systems, open standards for data contracts and software supply chains, and the catalog, exchange, and data-quality standards of the statistical community, alongside USAFacts' own later guidance. Many of the four qualities became tests a machine can run against an agency's website today, with no permission and no paid tools: whether the data is offered in a machine format, whether a documented machine interface and a full download exist, whether crawlers are allowed to reach it, whether it is listed for discovery and described in structured markup, whether it carries a machine-readable release date and a machine-readable licence, whether it is entered in a public inventory, and whether a plain account of how it was made can be reached from the data page. One of the four did not become such a test: accuracy asks whether a system that reads the data restates it correctly, which can be judged only by putting questions to each product and scoring the answers, not by reading a website, so it waits on funded tooling. A few finer points of the other qualities could not be made into a runnable test at all, for want of a settled method or a public signal to read, and these are marked plainly as open gaps rather than scored.

The result is that every one of those tests, the ones anyone can run from the open web, traces back to one of the framework's qualities or added areas, so a machine's real experience of an agency's website can be read in the very terms USAFacts set out, and measured the same way for all 16 federal statistical agencies.

Sources

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Derived from the project's readiness framework and its current public-web measurements across all 16 federal statistical agencies in September and October 2026. The detailed report holds the evidence behind each check.